

Vocab

Endogeneity

Exogeneity

Bias

Omitted variable bias

Internal and external validity

Consistency/inconsistency

Confounders

Mediators

Post treatment bias

Colliders and collider bias

Selection bias

- Sampling bias
- Survivorship bias
- Nonresponse bias

Measurement error

Simultaneity

Heteroskedasticity

Multicollinearity

Identification and causal identification

Instrumental variable

Two-stage least squares

Exclusion restriction

Regression discontinuity

- Continuity
- No manipulation

- Local randomization

- Local average treatment effect

Difference in differences

- Parallel trends
- Exogenous treatment assignment
- Anticipation effects
- Spillover effects

Theoretical and empirical estimand

Gauss Markov Theorem and assumptions

- Linearity
- Independence
- Homoscedasticity
- No perfect multicollinearity
- Exogeneity

Trial

Outcome

Event

Mutually exclusive events

Sample space

Random variable

Probability distribution

Probability density (mass) function

Cumulative density (mass) function

Normal and standard normal distribution

Parameter

Statistic

The law of large numbers

The central limit theorem

Sampling distribution

Sample distribution

Population distribution

Standard error

Standardized estimator

Z score

t-statistic

Standardized estimator

p-value

Confidence interval

Test statistic

Null hypothesis significance testing

t-test

Difference of means test

Permutation test

Non-normal residuals

Heteroskedasticity

Multicollinearity

Questions (second half of course)

What is endogeneity? Explain both conceptually and why the concept is significant.

How can we identify endogeneity?

How does R handle categorical variables when included as a control in a regression?

What is an interaction term? Why do we use them and how are they interpreted?

Sometimes it may be easier to interpret a model by looking at its predicted values rather than its coefficients. In what contexts would this be true?

What does it mean when we say a variable has been “absorbed” by the error term? What are the implications of this?

How does omitted variable bias/measurement error induce endogeneity?

Should you control for mediators in a regression model?

Explain three ways in which collider/selection bias can be introduced into a model.

What is heteroskedasticity/multicollinearity? What are the causes of it, how does it affect your model and its interpretation, and what can you do to resolve it?

Under what conditions is a multiple regression/instrumental variable/regression discontinuity/difference in differences model causally identified? Explain conceptually how the model isolates a treatment effect, and what assumptions identification rests on.

What is the law of large numbers and why does it matter?

What is the central limit theorem? Explain its significance in the context of regression analysis.

Explain the difference between a population, sample, and sampling distribution.

What is the standard error? How does it relate to the standard deviation?

When should you use a one-sided or two-sided hypothesis test?

What is the difference between a z-score and a t-statistic? Why would one be preferred over the other?

Why would you use a permutation test instead of a difference in means test?

Explain the significance of a model’s residuals in regression diagnostics.

Suppose you fit a regression model and examine the residuals. The residuals appear non-normal/heteroskedastic. Explain why this matters, how it might influence your analysis, and what you might do to diagnose and correct the issue.

What is multicollinearity and what causes it? Under what conditions does it matter, and under what conditions is it ignorable?

Explain all aspects of a regression table. Are the coefficients statistically significant? Under what conditions might we doubt the results of the hypothesis test? How might your interpretation of the coefficients, t-statistics, and R squared values change if your goal is causal inference vs. prediction?

What is IBE/abductive reasoning?

What is the difference between a theoretical and empirical estimand?

Why is it important to be able to think about a research question in terms of causal inference, even when estimating a causal effect is not your research objective?

Questions (first half of course)

What do we mean by the central tendency and how is it measured?

Suppose a random number is chosen from a distribution with the population mean of 10. Your best guess for that random number's value is 10. Why?

What assumption must hold for the difference in means estimator to be a valid estimator of a causal effect?

Suppose I want to estimate the causal effect of a treatment on a population. What assumption must hold for me to generalize the treatment effect in my experiment sample, to the population? How can this assumption be satisfied?

Explain all aspects of a regression equation.

What is the benefit of a simple random sample over a quota sample?

How does missing data undermine your estimation of a causal effect?

You are given a model and need to determine if a coefficient can be interpreted as a causal effect, and if the estimated effect is valid for a population, not just the sample studied.

What questions might you consider to make this determination?